Project Title

KrishiConnect: An Intelligent Platform for Agricultural Trade, Knowledge Sharing, and Market Insights

Project Problem Statement: Farmers often find it difficult to get reliable buyers, check current market prices, and get useful agricultural information at the right time. Different services are available on different platforms, so farmers may have to depend on several sources. KrishiConnect aims to solve this problem by bringing agricultural trading, knowledge sharing, market information, price forecasting, and intelligent recommendations into one platform.

Literature Review

The following research papers were studied to understand the existing work in agricultural marketplaces, digital agriculture, price prediction, recommendation systems, and AI-based market intelligence. The summaries below are written in simple, natural language while keeping the main ideas of the original sources.

Paper 1

Author & Year: Yash Suresh Rathore et al 2026

Title: Farmer to Farmer Marketplace

Methodology: This paper proposes a digital marketplace where farmers can directly buy, sell, exchange, or rent agricultural products and resources. The system includes registration and login, product management, search and filtering, communication, and transaction-related features. It also presents additional agricultural services such as weather information, disease detection, AI price prediction, and an AI chatbot.

Technology: The system uses a web/mobile application, database, user authentication, product search and filtering, communication tools, and digital agricultural services.

Findings: The proposed system makes it easier for farmers to find agricultural resources and communicate with other farmers without depending completely on intermediaries. It also shows how different agricultural services can be placed inside the same digital platform.

Limitations: The work mainly focuses on the marketplace and basic agricultural services. It does not provide a detailed integrated solution for advanced market intelligence, personalized recommendations, knowledge sharing, traceability, and large-scale real-world evaluation.

Relevance: This paper is especially useful for Krishi Connect because it gives a clear idea of how farmer-to-farmer buying, selling, exchanging, and communication can be supported digitally.

Research Gap: KrishiConnect can improve this concept by connecting the marketplace with price trends, forecasting, recommendations, ratings, notifications, knowledge sharing, and QR-based product traceability. This would make the platform more useful for both trading and decision-making.

Paper 2

Author & Year: Mohd Tajammul et al., 2025

Title: E-Commerce in Agriculture: A Review of Digital Marketplaces for Direct Farm-to-Consumer Trade

Methodology: The authors examined different digital agriculture and e-commerce platforms to know how technology can help farmers reach buyers directly. They looked at present marketplace models, supply-chain practices, and Instance from different countries. The review also considered practical problems such as infrastructure, digital skills, trust, and accessibility.

Technology: The study Talk about e-commerce platforms, mobile applications, AI, IoT, Big Data, predictive analytics, blockchain, online payments, GPS tracking, and multilingual services.

Findings: The study exhibits that digital marketplaces can give farmers improve access to markets and more transparent prices. Direct communication with buyers can also reduce dependence on middlemen and improve the efficiency of the agricultural supply chain. The review also shows that traceability and digital payments can make agricultural transactions more convenient.

Limitations: The paper is mainly a review of existing systems, so it does not develop and test one complete platform that combines all these services. Problems such as poor internet connectivity, low digital literacy, trust, cybersecurity, and rural infrastructure are still important barriers.

Relevance: This paper is closely connected to KrishiConnect because our project also aims to connect farmers and buyers through a digital marketplace. It supports the idea of online product listing, direct communication, transparent pricing, logistics, and traceability.

Research Gap: Most existing systems focus mainly on agricultural e-commerce. KrishiConnect can go a step further by bringing trading, farmer knowledge sharing, market-price analysis, price forecasting, recommendations, ratings, notifications, and QR-based traceability together in one platform.

Paper 3

Author & Year: Mobile App for Real-Time Market Information and Buyer-Seller Matching

Title: Mobile App for Real-Time Market Information and Buyer-Seller Matching

Methodology: The study presents a mobile marketplace designed to give farmers access to market information and help them find nearby buyers. The system was tested through functional, performance, and

usability evaluation. Essential functions such as registration, product uploading, product search, buyer-seller matching, and communication were checked.

Technology: The main technologies include a mobile application, database, search and filtering, location-based matching, communication, and notification features.

Findings: The testing showed that the main functions worked successfully. The reported response time was around 1–2 seconds, and the buyer-seller matching process achieved about 90% accuracy in the evaluation. The usability results were also positive, suggesting that a simple mobile interface can help farmers use digital market services.

Limitations: The system was evaluated mainly as a prototype using simulated usage situations. The authors suggest further improvements such as digital payments, AI-based market analysis, transport and logistics support, multilingual features, voice assistance, and testing with farmers in real agricultural environments.

Relevance: This work is directly relevant to KrishiConnect because finding buyers and providing market information are two important parts of our project. The location-based matching idea can also help farmers find suitable buyers or nearby opportunities.

Research Gap: The system mainly concentrates on market information and matching. KrishiConnect can extend it by adding price forecasting, knowledge sharing, product recommendations, ratings, notifications, and traceability.

Paper 4

Author & Year: Senthil Kumar V et al.

Title: FarmKart – E-Farm Connect: A Smart Agriculture Marketplace with Real-Time Delivery Tracking and Multi-Role System Architecture

Methodology: FarmKart presents a mobile-based agricultural marketplace with separate roles for consumers, farmers, delivery agents, and administrators. The system covers product browsing, farmer listings, inventory, orders, delivery tracking, payments, reviews, chat, coupons, invoices, and recommendations. The prototype was evaluated for functionality, performance, usability, and comparison with existing platforms.

Technology: The project uses Android Java/XML, Node.js, Express.js, REST APIs, Retrofit, MongoDB Atlas, Firebase, Google Maps, JWT authentication, payment integration, PDF generation, and real-time GPS tracking.

Findings: The system brings many parts of agricultural e-commerce into one application. It supports not only buying and selling but also delivery tracking, communication, reviews, invoices, and recommendations. The evaluation reported fast API response, quick notification delivery, and successful end-to-end testing.

Limitations: Since it is mainly a prototype, large-scale real-world deployment is not fully demonstrated. The future work includes multilingual support, AI/ML crop recommendations, demand forecasting, IoT-based cold-chain monitoring, cloud scaling, and blockchain-based product provenance.

Relevance: This paper is useful for KrishiConnect because it shows how a marketplace can be designed as a complete multi-user system instead of only a simple buying and selling website. Its delivery, review, notification, and recommendation features are especially useful references.

Research Gap: FarmKart focuses strongly on commerce and logistics. KrishiConnect can add a stronger intelligence layer by combining these functions with market insights, price prediction, farmer knowledge sharing, intelligent recommendations, and product traceability.

Paper 5

Author & Year: R. L. Manogna, Vijay Dharmaji, and S. Sarang, 2025

Title: Enhancing Agricultural Commodity Price Forecasting with Deep Learning

Methodology: The study compares several statistical, machine-learning, and deep-learning methods for predicting agricultural commodity prices. It uses long-term daily price data from multiple agricultural markets and evaluates the models using forecasting error measures. The study mainly focuses on how well different models can capture changing and nonlinear price patterns.

Technology: The researchers compare ARIMA, Support Vector Regression, XGBoost, MLP, RNN, LSTM, GRU, and Echo State Networks.

Findings: The results show that deep-learning models such as LSTM and GRU can perform well when agricultural prices contain complex time-based patterns. The study also points out that weather and other external information could make future forecasting systems more useful.

Limitations: The model performance depends on the available historical data and the markets covered by the study. Sudden events such as policy changes, extreme weather, or other unexpected market shocks may not be fully captured by historical price data alone.

Relevance: This paper is important for KrishiConnect because price forecasting is one of the main market-intelligence features planned for the platform. It gives useful evidence for considering time-series and deep-learning models.

Research Gap: The study focuses mainly on forecasting and does not provide a complete agricultural trading platform. KrishiConnect can connect the forecasting model directly with farmer-buyer trading, market information, recommendations, and decision support.

Paper 6

Author & Year: P. Ankit Krishna, Gurugubelli V. S. Narayana, Siva Krishna Kotha, and Debabrata Pattnayak, 2026

Title: Machine Learning Based Agricultural Price Forecasting for Major Food Crops in India Using Environmental and Economic Factors

Methodology: The authors developed a machine-learning system for predicting crop prices using more than just previous prices. Their dataset includes environmental, economic, market, and transportation-related factors. They followed steps such as data collection, preprocessing, feature engineering, feature selection, model training, and evaluation.

Technology: The study tests Linear Regression, Support Vector Machines, Random Forest, AdaBoost, and XGBoost. It also uses feature-selection and data-preprocessing techniques.

Findings: XGBoost gave the best performance among the tested models, with a reported R^2 of 0.94 in the paper. The study also found that factors such as temperature, rainfall, demand and supply, fertilizer use, transportation cost, and regional competition can influence agricultural prices.

Limitations: The study uses a particular dataset and time period, so the model may not perform equally well in every region or during unusual market situations. Sudden policy changes, extreme weather, and other unexpected events are difficult to represent using only the available historical variables.

Relevance: This paper is highly relevant to KrishiConnect because it shows that price prediction can be improved by considering environmental and economic factors rather than depending only on past prices.

Research Gap: KrishiConnect can use this idea in a farmer-friendly decision-support module. Instead of showing only a predicted price, the system can combine price forecasts with current market conditions, trading opportunities, recommendations, and agricultural knowledge.

Paper 7

Author & Year: Bryan C. Runck et al., 2021/2022

Title: Digital Agriculture Platforms: Driving Data-Enabled Agricultural Innovation in a World Fraught with Privacy and Security Concerns

Methodology: This paper looks at digital agriculture platforms from a broader perspective. The authors classify different types of platforms and discuss how agricultural data can be collected, shared, processed, and reused. They also examine the challenges involved in managing agricultural data across different users and systems.

Technology: The paper discusses digital agriculture platforms, data repositories, analytics, dashboards, data sharing, interoperability, security, privacy, and data-management technologies.

Findings: The study explains that digital platforms can support data-driven innovation in agriculture by bringing different kinds of information and services together. At the same time, it highlights concerns about data quality, ownership, interoperability, privacy, security, and access.

Limitations: Digital agriculture is a rapidly changing field, so there is no single universally accepted definition or classification of a digital agriculture platform. The work is therefore more of a conceptual framework than a complete farmer-facing application.

Relevance: This paper is useful for KrishiConnect because our project will also need to combine different types of agricultural data and services. It reminds us of that data security, privacy, interoperability, and good data management are important while designing the platform.

Research Gap: The paper provides a broad platform framework but does not focus specifically on an integrated farmer marketplace. KrishiConnect can apply these data-management ideas while building trading, market intelligence, knowledge sharing, and recommendation features.

Paper 8

Author & Year: Varun Prasanna M et al., 2026

Title: Digital Platforms in Agricultural Marketing: A Case Study of Accessibility, Scope and Constraints in Chikkaballapur District, Karnataka

Methodology: The study collected primary information from 120 farmers in Chikkaballapur and examined how they use digital platforms for agricultural marketing. A mixed-method approach was used to understand platform usage, benefits, and problems. The researchers studied platforms such as e-NAM, UMP, social media, Krishi Bandhu, and e-Samridhi.

Technology: The study examines digital marketing platforms including e-NAM, UMP, social media, Krishi Bandhu, and e-Samridhi, along with mobile and online digital services.

Findings: The study found that farmers value real-time market prices, direct farmer-to-buyer transactions, and better market access. UMP was reported as the most regularly used platform in the study. At the same time, unreliable internet, power cuts, poor digital infrastructure, and lack of devices were major challenges.

Limitations: The research is limited to Chikkaballapur and a sample of 120 farmers, so the results may not represent every farming region. It studies the use and perception of digital platforms rather than developing a completely intelligent marketplace.

Relevance: This is particularly useful for KrishiConnect because it provides evidence from Karnataka itself. It shows that farmers are interested in timely price information and direct access to buyers, while also reminding us that the system must be simple and usable in areas with connectivity limitations.

Research Gap: The existing platforms studied in the paper provide different services separately. KrishiConnect can reduce this fragmentation by combining market information, trading, forecasting, knowledge sharing, recommendations, and notifications in one place.

Paper 9

Author & Year: Ramaraj R, Karthick Raja R, Mathivasan S P, Sakthisivabalaji P, and Santhosh K, 2026

Title: Development of a Smart Agricultural Marketplace with Machine Learning-Based Price Forecasting

Methodology: This research combines an online farmer-buyer marketplace with a machine-learning model for crop-price prediction. Historical market data is cleaned and prepared, useful features are extracted, models are trained and compared, and the selected model is connected to a Flask web application where farmers can view predicted prices.

Technology: The system uses Python, Flask, HTML, CSS, JavaScript, SQLite/MySQL, Scikit-learn, Pandas, NumPy, Matplotlib, Random Forest, and Linear Regression.

Findings: The study reports that Random Forest performed much better than the Linear Regression baseline. The proposed platform allows farmers to obtain estimated prices and interact with buyers. The paper therefore demonstrates the practical benefit of putting price predictions and digital trading together.

Limitations: The authors identify problems such as incomplete or noisy data, differences between regions, scalability, limited digital literacy, and data security. They also suggest future additions such as real-time market feeds, mobile deployment, secure transactions, and regional price comparison dashboards.

Relevance: This is one of the closest papers to KrishiConnect because it combines the two core ideas of our project: agricultural marketplace functionality and machine-learning-based price forecasting.

Research Gap: The system mainly connects trading and price prediction. KrishiConnect can make the idea broader by adding knowledge sharing, personalized recommendations, ratings, notifications, buyer matching, and QR-based traceability.

Paper 10

Author & Year: Abhishek Kumar, 2026

Title: AI-Powered Local Agri-Market Intelligence and Farmer-Buyer Connect System

Methodology: This work develops an AI-based platform that helps farmers understand local agricultural markets and connect with buyers. It uses multiple AI agents for tasks such as understanding user questions, identifying locations, retrieving market information, checking prices, and generating responses. The system also uses retrieval-based methods so that answers can be connected to relevant agricultural and government information.

Technology: The system uses multi-agent architecture, RAG, FAISS, sentence-transformer embeddings, agricultural market data, web search, React, Tailwind CSS, Node.js, Express.js, MongoDB, Python, FastAPI, and cloud deployment tools.

Findings: The platform demonstrates how conversational AI can make market information easier for farmers to understand. The reported evaluation includes high intent-classification and price-verification performance, along with positive farmer satisfaction scores. The bilingual approach also makes the system more accessible.

Limitations: The study reports limitations in speech recognition for some regional accents and difficulties with ambiguous queries. Market data can also be delayed, and the price-analysis component does not fully model complex nonlinear time-series behaviour or sudden market shocks.

Relevance: This paper is highly relevant to KrishiConnect because it shows how AI can be placed between complex market data and farmers, giving them simpler explanations and useful recommendations rather than only raw numbers.

Research Gap: The system focuses more on market intelligence and conversational support than on a complete transaction ecosystem. KrishiConnect can combine this type of AI assistant with product listings, buying and selling, knowledge sharing, recommendations, ratings, notifications, and traceability.

Overall Research Gap for KrishiConnect

From the reviewed papers, it is clear that researchers have already developed useful solutions for individual agricultural problems. Some systems focus on buying and selling, some provide market prices and buyer matching, some predict crop prices, while others provide AI advice, recommendations, coordination, or traceability. However, these functions are usually separated rather than being offered as one complete platform.

KrishiConnect aims to bring these ideas together in a single farmer-friendly system. The platform can combine direct farmer-buyer trading with real-time market information, price forecasting, knowledge sharing, personalized recommendations, ratings, notifications, and QR-based traceability. This makes the project different from a simple agricultural marketplace because it is designed to support both trading and better decision-making.